

Ch. 8 — P03 (D) (s,S) Policy Optimization

Problem Bank | Chapter 8 | Type: D | Difficulty:

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Scenario

A regional distribution center manages inventory for a single SKU under a continuous-review (s, S) policy.

Parameter	Value
Daily demand	Poisson with mean $\bar{d} = 8$ units/day
Lead time	Uniform[2, 5] days
Unit holding cost	\$0.50/unit/day
Fixed ordering cost	\$100 per order
Unit shortage cost	\$5/unit/day of shortage
Review period	Continuous
Simulation horizon	365 days

The current policy is $(s, S) = (20, 60)$.

Tasks

- Baseline simulation:** Using `simdes.models.inventory.SSInventory` (or your own SimPy implementation), run 30 replications of the current $(20, 60)$ policy. Report the mean daily cost decomposed into holding, ordering, and shortage components, each with a 95% CI.
- Parameter sensitivity:** Fix $S = 60$ and vary $s \in \{5, 10, 15, 20, 25, 30\}$. For each value of s , run 20 replications and plot mean daily holding cost, ordering cost, and shortage cost on the same figure (three lines). Identify the tradeoff: as s increases, which cost goes up and which goes down?
- Grid search:** Search over $s \in \{10, 15, 20, 25, 30\}$ and $S \in \{40, 50, 60, 70, 80\}$ with the constraint $S > s + 10$. Use 30 replications per policy. Present results as a heatmap of mean daily total cost. Mark the empirically optimal policy with a star.
- Common Random Numbers:** Using CRN (same seed sequence for each replication), compare the best policy from Task 3 against the current $(20, 60)$ policy using 50 paired replications. Report the mean cost difference and its 95% CI. Is the difference statistically significant?

5. **Robustness check:** Re-run your optimal policy under two demand perturbations:

- (a) Demand rate doubled ($\bar{d} = 16$)
- (b) Lead time extended (Uniform[4, 8] days)

Does your policy remain near-optimal, or does it perform significantly worse? Recommend how to adjust (s, S) for each perturbation.

6. **Reflection:** In 150 words or fewer, explain why the grid search is not a practical approach for a real distribution center that manages thousands of SKUs. What would you do instead? (This sets up Module M12.)

Deliverables

- A Jupyter notebook with all simulation code, plots, and CRN comparison
- The heatmap from Task 3 (publication quality)
- A one-page management summary stating the recommended (s, S) policy and expected cost savings versus the current policy